Cointegration and the VECM Model

Chapter 8

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Outline

- Why differencing is not good enough
- Cointegration
- The cointegration test
- The Johansen trace test
- The VECM Model
- VECM diagnosis
- IRFs and the VECM forecast

Why differencing is not enough

Section 1

Why differencing is not enough

- First, a recap
  - We started looking at one time-series variable
    - Is it stationary? If it is not, what kind (trend, unit-root)?
  - We learned how to model stationary and non-stationary series
  - We move to a more realistic scenario for economics: More than one time-series variable with endogeneity
  - What if more than one series is non-stationary?
    - Using levels: Wrong
    - Differencing: Might throw away useful information
    - If the non-stationary series are somehow related, we lose that information by differentiating everything

Why differencing is not enough

- Example
  - The 3M T-bill and the 10Y T-bill do not move perfectly in sync, but they do not drift apart permanently — they are somehow tied to each other
  - Remember the expectations hypothesis of the term structure
    - The long-run interest rate equals the average of short-term rates (plus a constant premium): $i_{LT} \approx \sum_j i_{j,ST}$
    - This means both rates share the same underlying stochastic trend, and their **spread should be stationary**

Why differencing is not enough

- VAR in levels
  - With non-stationary series, the regression and inference fail
- VAR in differences
  - The spread (stationary) among two or more I(1) series is lost
  - This is omitted information
- Solution
  - Bring cointegration to the regression

Cointegration

Section 2

Cointegration

- Cointegration
  - A stationary combination of two I(1) series
  - Assume two non-stationary stochastic series that share a trend
    - $y_t = \bar{y} + \tau_t + \varepsilon_t^y$
    - $x_t = \bar{x} + \tau_t + \varepsilon_t^x$
    - With $\tau_t = \tau_{t-1} + \eta_T$
    - $\varepsilon_t^y \sim N(0, \sigma^y), \varepsilon_t^x \sim N(0, \sigma^x), \varepsilon_t^\eta \sim N(0, \sigma^\eta)$
  - Now take the difference and make the trend disappear
    - $y_t - x_t = (\bar{y} - \bar{x}) + \left(\varepsilon_t^y - \varepsilon_t^x\right)$
    - $y_t - x_t = \text{constant} + \text{difference of two stationary processes}$
    - $y_t - x_t$ is stationary

Cointegration

- Formal definition
  - Two I(1) series y_t, x_y are cointegrated if there exists a constant β such that $z_t = y_t - \beta x_t$ is I(0). $(1, -\beta)'$ is the **cointegrating vector**. z_t is the equilibrium residual or **error correction term**

Cointegration

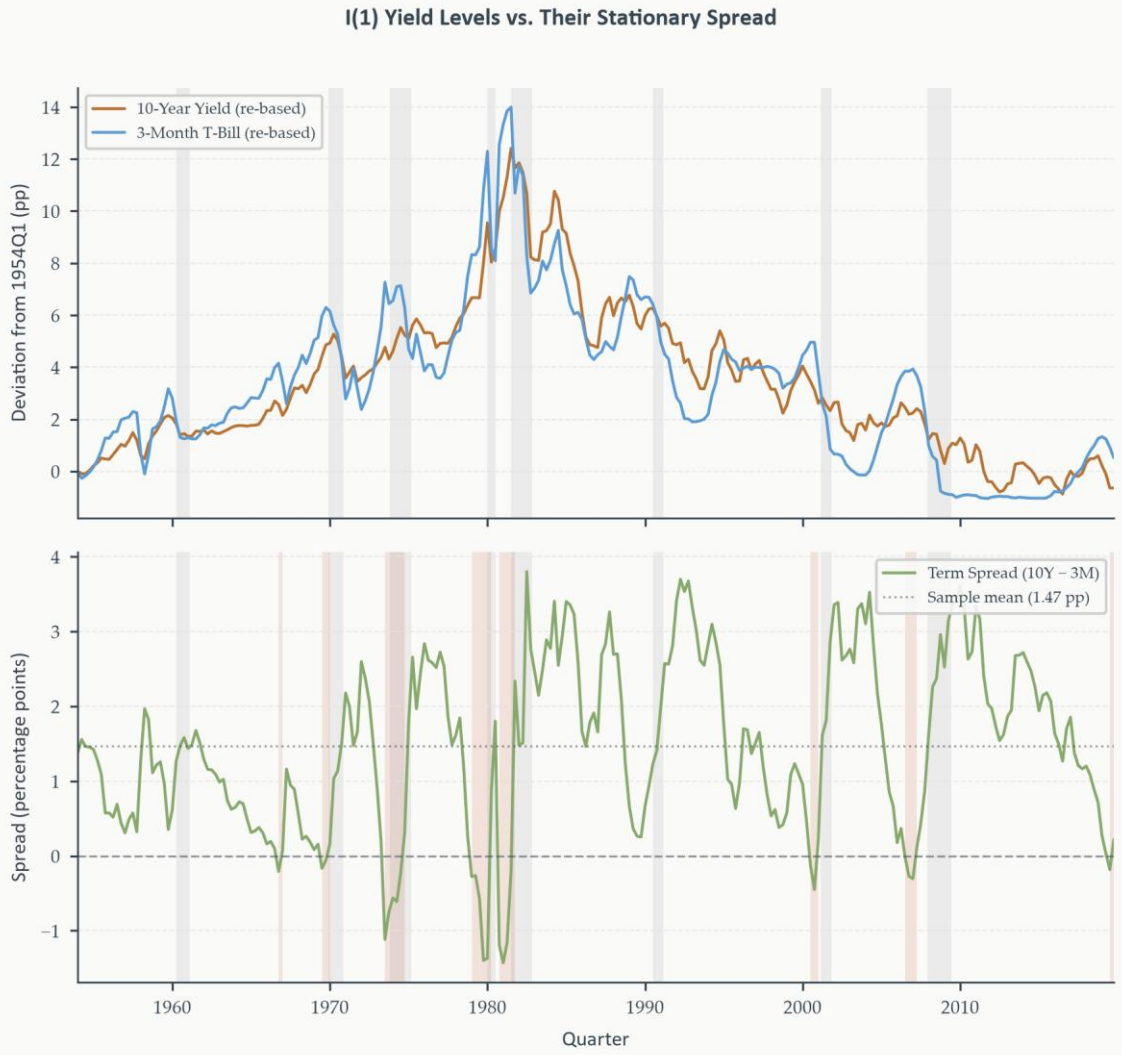
- If $z_t = y_t - \beta x_t$ has mean $\bar{z}$, then in the long run (when z_t sits at its mean):
  - $y_t = \bar{z} + \beta x_t$
  - β is the long run response of y_t to a **permanent** change in x_t
  - The equilibrium residual measures the current deviation from that relationship
    - Positive (negative) means y_t sits above (below) its long-run value given x_t
  - For the yield curve example, $i_t^{10Y} - i_t^{3M}$ is stationary around a constant term $\bar{z}$ (the expected hypothesis prediction)

Cointegration

- Could there be more than one independent long-run relationship at once?
  - That's the cointegration rank r: The number of linearly independent cointegrating vectors among n $I(1)$ variables, such that $0 \leq r \leq n-1$
  - $r = 0$: no cointegration, n independent trends → difference your VAR model
  - $r > 0$: cointegration (r equilibrium relationships), and $n - r$ independent trends → run a VEM model
  - $r = n$: Impossible (it would mean all variables are stationary)

Cointegration

- Example
  - ADF and KPSS point to both i_t^{3M} and i_t^{10Y} being stationary processes
  - But their spread is bounded



The Johansen trace test

Section 3

The Johansen trace test

- Start from a VAR(p) in levels
  - $y_t = c + A_1 y_{t-1} + \cdots + A_p y_{t-p} + \tilde{\varepsilon}_t$
  - Subtract y_{t-1} from both sides and collect terms (similar to the ADF test construction)
  - $\Delta y_t = c + \Pi y_{t-1} + \Gamma_1 \Delta y_{t-1} + \cdots + \Gamma_{p-1} y_{p-1} + \xi_t$
  - $\Delta y_t = \left(c + \Gamma_1 \Delta y_{t-1} + \cdots + \Gamma_{p-1} y_{p-1}\right) + \Pi y_{t-1} + \xi_t$
    - $\sum_{i=1}^{p} A_i - I_n \,, \quad \Gamma_i = -\sum_{j=i+1}^{p} A_j$
    - Looks like a VAR in differences with an extra term
    - Γ_i capture the short-run dynamics
    - Πy_{t-1} is the only term in levels, capturing the long-run information

The Johansen trace test

- Now, look at Π
  - $rank(\Pi) = 0 \rightarrow$ no level terms; the differenced VAR is the correct model
  - $rank(\Pi) = n \rightarrow$ full rank, meaning all variables are stationary in levels
  - $0 < rank(\Pi) < n \rightarrow$ VECM is the correct model
- Matrix Π is unobserved, we must work with an estimate: $\widehat{\Pi}$

The Johansen trace test

- Trace statistic
  - $H_0: rank(\Pi) \leq r$
  - $H_1: rank(\Pi) > r$
  - $\lambda_{trace}(r) = -T \sum_{i=r+1}^{n} \ln\left(1 - \hat{\lambda}_i\right)$
- Maximum eigenvalue test
  - $H_0: rank(\Pi) = r$
  - $H_1: rank(\Pi) = r + 1$
  - $\lambda_{max}(r, r+1) = -T \cdot ln\left(1 - \hat{\lambda}_{r+1}\right)$

The Johansen trace test

- Sequential testing
  - Start with the most restrictive scenario
    - Test $r = 0$ against $r \geq 1$
    - If reject, move to $r = 1$ against $r \geq 2$
    - Continue until first failing to reject: That's is rank $\hat{r}$

H_0: rank $\leq$	Eigenvalue	Trace	CV 95%	Decision
0	0.107	33.33	15.49	Reject
1	0.014	3.61	3.84	Fail to reject
Estimated rank $\hat{r} = 1$ One cointegration relationship (one common stochastic trend)				

The VECM model

Section 4

The VECM model

- $\Delta y_t = c + \Gamma_1 \Delta y_{t-1} + \cdots + \Gamma_{p-1} y_{p-1} + \Pi y_{t-1} + \xi_t$
- Matrix $\Pi = \alpha\beta'$ → what's this?
- Using the yield curve example
  - $\Delta i_t^{3M} = \bar{\imath}_{3M} + \alpha_1 \underbrace{(i_{t-1}^{10Y} - \beta r_{t-1}^{3M})}_{error\ correction} + \gamma_{11} \Delta i_{t-1}^{3M} + \gamma_{12} \Delta i_{t-1}^{10Y} + \xi_{1t}$
  - $\Delta i_t^{10Y} = \bar{\imath}_{10Y} + \alpha_2 \underbrace{(i_{t-1}^{10Y} - \beta r_{t-1}^{3M})}_{error\ correction} + \gamma_{21} \Delta i_{t-1}^{3M} + \gamma_{22} \Delta i_{t-1}^{10Y} + \xi_{2t}$
  - EC signs
    - Positive: Long-run rate above the short-run rate
    - Negative: Inverted yield curve

The VECM model

- Example (realistic output)

	Coefficient	Interpretation
$\hat{\beta}$ (3M rate)	1.0218	Long-run one-for-one comovement
Term premium	≈ 1.35pp	Matches sample mean spread (1.47pp)
$\hat{\alpha}_1$ (3M yield)	0.084**	Short rate rises when spread above equilibrium
$\hat{\alpha}_2$ (10Y yield)	-0.056**	Long rate falls when spread above equilibrium

VECM diagnosis and common pitfalls

- A VECM model is correctly specified if residuals are
  - White noise
  - Serially uncorrelated
  - Homoskedastic
  - Approximately normal
- What happens if you forget the EC term?
  - The residual ACF is almost the same between a VAR(p) and VECM(p)
  - EC contribution is “invisible”: It is about long-term stability

VECM diagnosis and common pitfalls

- Common mistakes
  - Pitfall 1: Over-differencing
    - EC accumulates as forecast error over “long” horizons
  - Pitfall 2: Spurious cointegration
    - Be careful of wrongly rejecting “no cointegration”
    - What if cointegration does not have strong theoretical support?
  - Pitfall 3: Rank Misspecification
    - If rank is too low: Omits EC terms
    - If rank is too high: Introduces spurious EC terms (adds noise without explanatory power)
    - Estimate VECM at adjacent ranks and compare their diagnostics

VECM diagnosis and common pitfalls

- Be careful
  - Like most models, VECM does not automatically account for structural breaks
  - Tet for cointegration within stable subsamples, use dummy variables

VECM diagnosis

Section 5

VECM diagnosis

- VECM diagnosis
  - The usual:
    - Serial correlation: Ljung-Box test
    - Normality of residuals: Jarque-Bera
  - Remember you have one set of diagnostic tests per equation

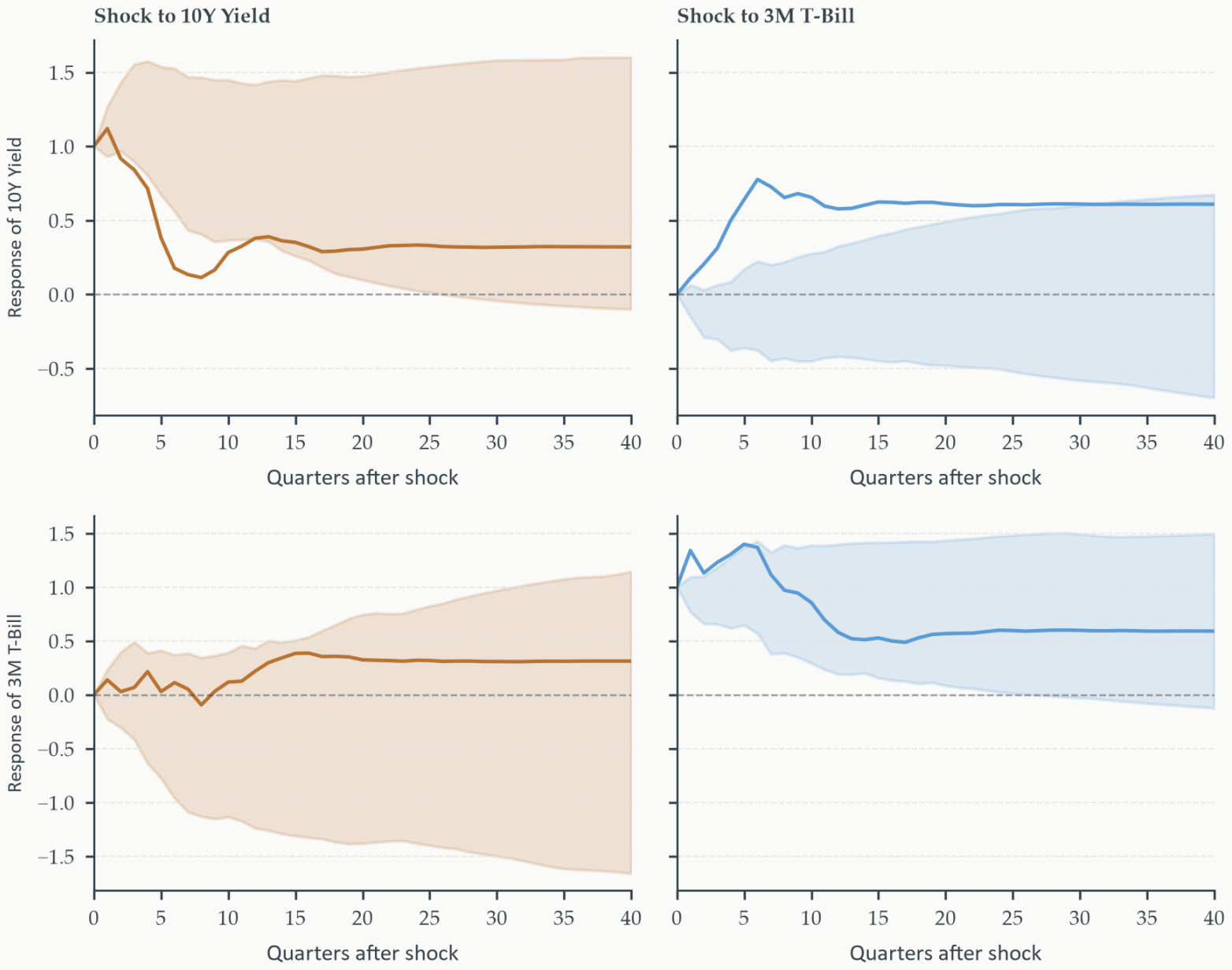
IRFs and the VECM forecast

Section 6

IRFs and VECM forecast

- The VECM model is an “expanded” VAR, therefore, estimation follows similar steps to a VAR model
  - Lag selection
  - Cholesky decomposition
  - Model stability
- What’s different?
  - In a stationary VAR, every IRF converges to zero (transitory shocks)
  - In a VECM with $r = 1$, responses converge to a **nonzero permanent level** instead

VECM Impulse Responses — Yield Curve System



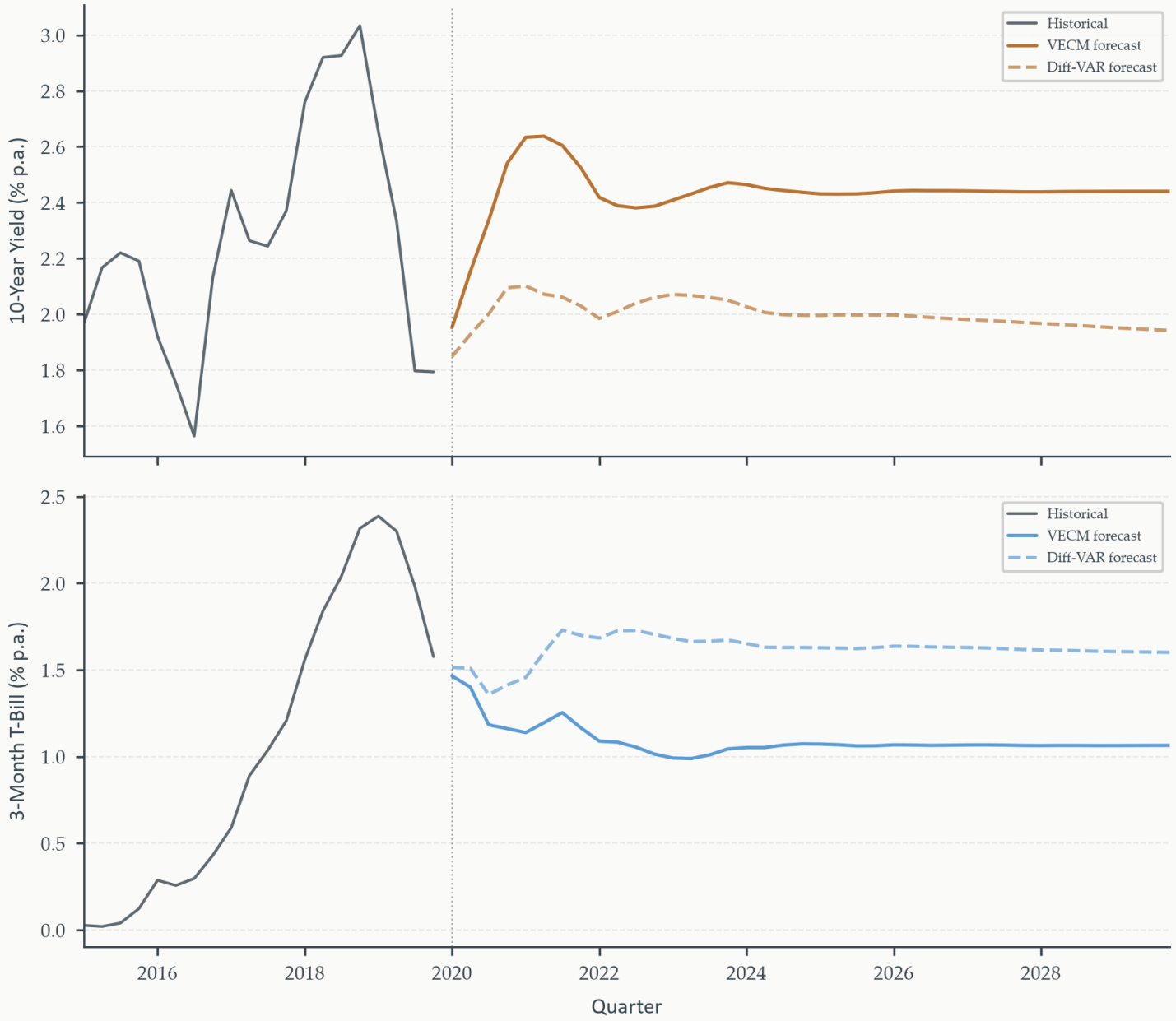
IRFs and VECM forecast

- Why responses never return to zero?
  - Every shock has two components
    - A **permanent** shock moves the common stochastic trend (equilibrium) itself
    - A **transitory** shock moves the spread away from equilibrium

IRFs and VECM forecast

- Consider the following two scenarios
  - Volcker disinflation: a **permanent** shock that raises short-run rates more than long-run rates
  - Flight-to-quality: A **transitory** shock that can produce an inverted yield curve
  - Neither VAR in levels nor VAR in differences can capture the distinction
  - The VECM model can separate short-run from long-run effects

VECM vs. Differenced VAR: 40-Quarter Forecasts



IRFs and VECM forecast

- Implications
  - For long-run forecast, VECM is the correct specification; a VAR model will suffer from omitted variable bias
    - Pension fund liability matching
    - Monetary policy projections
    - Fixed-income portfolio construction

Cointegration and the VECM Model

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